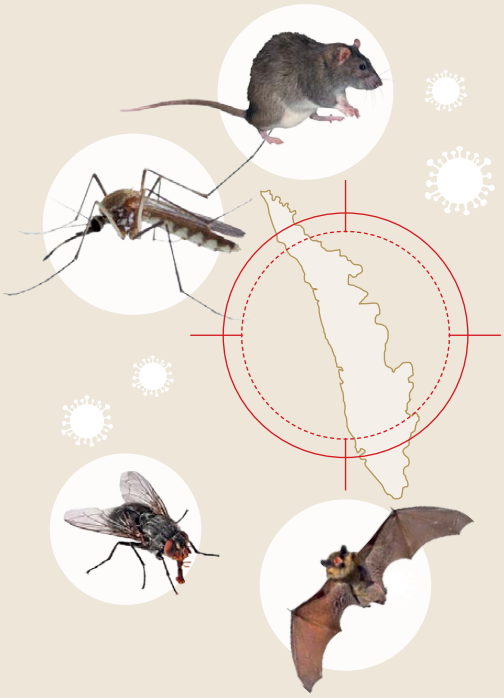
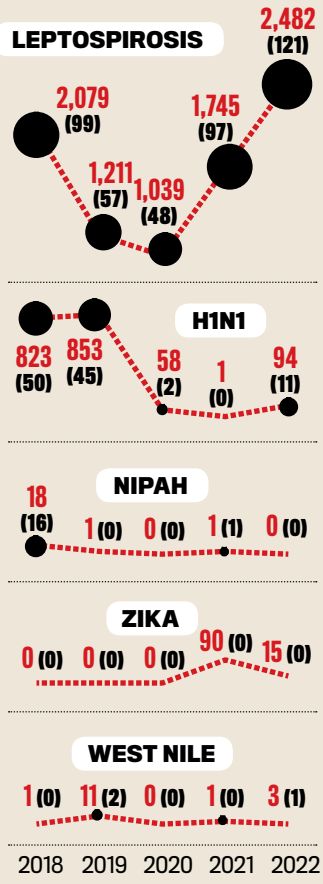
OUTBREAKS TAKE A TOLL

Kerala has become the epicentre of deadly vector-borne and zoonotic infections in recent years





Source: Kerala Government

KERALA

GOD'S OWN VIRUS LAB

in Kozhikode district confirmed they had contracted Nipah. As of September 18, four infected persons were still hospitalised while the health department had identified and isolated 1,233 persons who had come in contact with them or the index patients, including 352 said to be in the high-risk category. "So far, tests conducted on high-risk contacts have returned negative and no new case has been reported," says health minister Veena George, who is camping in Kozhikode. "Even the adjoining districts of Malappuram, Wayanad and Kannur have not notified any symptoms." The minister is hopeful of having contained the virus, but says the department will remain on alert for another three weeks to prevent a second wave.

Kerala had first reported Nipah in 2018, in Kozhikode and Malappuram. It was brought under control, but only after 16 of 18 infected patients had succumbed—a mortality rate far worse than, say, Covid's. Cases resurfaced in Kochi in 2019 and again in Kozhikode in 2021. Chikungunya, spread by the bite of a mosquito, too, keeps resurfacing in the state: 66 cases were reported last year. Although it has caused no death in Kerala in the past decade, the dengue virus—also spread by mosquitoes—has not been that kind. It killed as many as 58 people in 2022. This year, till July, the state had recorded 3,409 confirmed cases of infection. Leptospirosis, a bacterial infection transmitted by contact with the urine of rodents and farm animals or contaminated soil or water, also has a high incidence in the state—2,482 confirmed cases and 121 deaths, with another 2,833 suspected cases and 169 deaths, in 2022. Despite a wide network of public health institutions, this rising number of contagious diseases has become a headache for the state's policymakers and medical personnel.

So why has Kerala become an epicentre of such diseases? "Studies in infected areas have revealed that most cities, coastal areas and hilly regions in the state do not have a proper drainage system," says Dr T.G. Vinodkumar, scientist at the Jawaharlal Nehru Tropical Botanic Garden and Research Institute, Palode. So everytime it rains, stagnant water becomes a breeding ground for disease-spreading mosquitoes. "After the floods in 2018, even the number of leptospirosis cases almost doubled," he says. Add to that a high population density, with about 1.5 million migrant workers from other states residing in Kerala, and another 2.3 million Keralites working abroad, many of whom would travel back home on any given day, it's not easy to keep tabs on those carrying infections—Zika, West Nile, and such.

The Indian Council of Medical Research and the National Institute of Virology, Pune, in a study conducted in 16 states/ Union Territories in July, had found the presence of Nipah virus in the bat population of 10 states/ UTs, including Kerala. Why is it that just Kerala has reported infections, yet again? The health minister puts it down to its better health reporting and surveillance systems. "We account for all infections. Many states don't do such monitoring," says George. "But it's true that we have to clean up our drains, wells and canals during and after monsoons to check mosquitoes." That may well be the missing piece in the puzzle...or not. ■